

The WD Raptor has a random access time of 6 ms, matched only by the Hitachi 1 TB drive. Other drives fall far behind here, with double-digit random access times. We also see this performance in the file copy tests. The Green Power drives from WD posted the worst access time, thanks mostly to the slower rotation speed.

The WD Caviar SE16 320 GB, Seagate 7200.11 drives and the single platter Seagate 250 GB sported the best Average Read /

Write speeds. The large 16 and 32 MB caches are doing their bit to speed up the read and write operations of these drives. Other drives were not far behind, but a difference of 10 Mbps between hard drives translates to a lot more in practical usage. Seagate's Barracuda 7200.10 250 GB marches ahead of other drives in this test, thanks to a perfect balance of cache, interface speed and single platter configuration.

H2Bench
Another low-level benchmark which reproduces precise results and requires raw hard drives—that is, drives without any file system or partitions. Seagate's Barracuda 7200.11 posted the best sequential transfer speeds—we attribute this to better drive electronics and higher cache. The WD 320 GB AAKS (last year's winner), still holds its ground and delivers a performance as

How We Tested

The performance test for hard drives consists of synthetic tests, benchmarking software and real-world tests—where application performance and file transfer speeds are tested. The synthetic tests were conducted using two types of benchmarks—low-level benchmarks—namely, HD Tach RW 3.0.1.0 and H2Bench, and standard benchmarks PCMark 05 and SiSoft Sandra XI SP1.

Features
In the features, we noted and rated various features of the hard drive such as capacity, dimension, weight, bundled software, interface type, cables and accessories and ruggedness.

Performance
HD Tach RW 3.0.1.0 was used for noting down the following parameters in the test hard drive.

CPU Utilisation
Lower CPU utilisation is better, and is measured as a percentage of the maximum.

Random Access Time Benchmark
Random access is the time required to retrieve data from random locations on the drive. It is represented as time—the lower the value, the better the drive.

Average Read And Write Speed
It is the speed with which data can be accessed from the drive. The speed is represented as throughput and generally specified as Mbps—the higher the score, the better the drive.

Burst Speed
The maximum interface speed that can be achieved by the drive is known as the burst speed. This is identical to PMPO in speakers. Burst speeds are indicated in terms of Mbps—the higher the score, the better the drive.

H2Bench
With H2bench, we measure the interface speed at 50 per cent of the drive's capacity. The "Core Test" or Repetitive Sequential Read Test was then performed and through repeated sequential reads, revealed the maximum transfer rate of the drive. The read and write seek times were also noted.

PCMark 05
FutureMark's PCMark 05, is a very popular tool to benchmark the entire system. We noted the scores of the following components from the hard drive benchmark module.

XP Startup: This simulates a typical Windows XP startup, and tests the performance of the hard drive at initiation.

Application Loading: This simulates the opening and closing

Test Rig Configuration	
Processor	Intel Core 2 Extreme X6800
Motherboard	Intel D975XBX2
Chipset	Intel 975
Memory	1 GB Corsair at 800 MHz
Host Hard Drive	Hitachi 500 GB
OS	Windows XP SP2 (updated)

of several applications such as Adobe Reader and Windows Media Player.

Virus Scan: This tests the drive performance during a simulated virus scan of approximately 600 MB of files with different file types.

SiSoft Sandra XI SP1
The scores of Sequential Read, Random Read, Sequential Write, Random Write and Access Time were reported at the end of this benchmark, which we duly noted. This is one benchmark that reproduces constant results, which is why we still use it.

Real World Tests—File Copy
The Real World tests consisted of copying a 4 GB (1 GB for external drives) file (for sequential data transfer speed) and 4 GB of assorted files (for random data transfer speed). The time taken for this data transfer was noted down. The assorted files consisted of multiple file types that included applications, Word documents, Excel files, MP3s and small video clips, thus simulating the real world situation where such files will be copied.

Three tests were carried out—first Read/Write Assorted files, followed by Read / Write sequential file and lastly, both file types were copied from one partition of the test hard drive to the other. The last test puts the hard drive to the highest stress levels and gives a clear verdict of how good the drive performs in a real world scenario, where we end up moving data between partitions.

Real World Test—Photoshop CS3
We installed Adobe Photoshop CS3, and configured the scratch disk to occupy the first partition of the test drive. We then started the application and noted the time taken for it to open completely. We then copied two PSD files, a 550 MB and a 1 GB, to the first partition of the test drive and clocked the times taken to open each of these.

good as Seagate's new generation drives. Seagate's Barracuda 7200.11 500 GB trumps in this test. Hitachi drives, especially the 1 TB does come really close to the winner, but loses out by a tiny margin. Though one would expect the WD Raptor to deliver a knockout performance, its 1.5 Gbps interface limits it. We expect the new VelociRaptor to put up a better show.

PCMark 05
The well known system-wide test suite, PCMark 05, has a special test aimed at gauging hard drive performance. In the XP start up sub-test, WD's Raptor scored the highest. This is largely due to its low access time (seek time). Similar results are seen with the Hitachi 1 TB drive that has a low access time of 6 ms. Seagate's 7200.11 and the WD Caviar SE16 750GB followed closely. In the Virus scan test—important because it involves lot of IO—the Seagate Barracuda 7200.11 500 GB, delivered an astonishing 142 Mbps. Seagate's 750 GB and 1 TB were trailing, but still ahead of the WD and Hitachi drives. Seagate's 80 and 160 GB Barracuda are at the bottom of the heap and it's partly the 2-MB cache that's to blame! Please stay away from these drives—the 250 GB is a much better bet.

SiSoft Sandra XI SP1
Seagate's Barracuda 7200.11 500 GB posted the highest Drive Index score, followed very closely by Western Digital's Caviar 320 GB and Seagate's 7200.11 750 GB. These three drives have significantly better sequential read and write speeds, and they overshadow other drives. Despite lower interface speeds, the WD Raptor does deliver when it comes to random read / write operation—better access times make up for the lower interface speeds. WD's RE2 750 GB posted better scores



Barracuda 7200.11 500 GB

across the various sub-tests and is one of the better drives other



Seagate's Barracuda 7200.11 750 GB

than the Seagate 7200.11 drives. Overall, the Seagate Barracuda 7200.11 500 GB came out as the winner.

Real World Copy Tests
In the assorted write test, the Western Digital Raptor completed the test in the least possible time. The 10K spindle speed coupled with low access times helps it achieve this feat. Western Digital Caviar 320 GB and Seagate's 7200.11 500 GB were next set of drives to complete the file transfer. All other drives took 4 to 5 seconds more than these two drives to complete the test. The 80 GB and 160

GB from Seagate took 43 seconds more, and were way behind all drives. These two drives are seriously handicapped and should be avoided.

Almost all drives did well on the Assorted Read test, and delivered consistent results, taking approximately 83 seconds to read 4 GB of data. In the partition-to-partition file copy test, WD's caviar 320GB completed the feat in the least time. In this test, the buffer memory makes all the difference. However, apart from buffer size, buffer speed also dictates the final outcome. For example, the Seagate's 7200.11 has a 32-MB buffer. However, these are outdone by the 7200.10 Seagate drives and we presume it is due to the buffer speed.

The sequential read / write test results are similar to the assorted read / write test results. However, in sequential partition-to-partition file transfer, the Western Digital Caviar 320 GB takes the pole position dethroning Raptor. The other drives put up reasonably good performance, barring the two Seagate's namely 80 and 160 GB.

Winners
Seagate's Barracuda 7200.11 500 GB delivered consistent performance throughout, and we adjudge it as the Best Performer. Western Digital's Caviar 320 GB (our last year's winner), still has enough steam to stay ahead of the competitors. However, given the price difference, we recommend Seagate's 500 GB. You get 160 GB of more space at a paltry difference of just 400 bucks. For its superb value for money, we also award the Seagate Barracuda 7200.11 500 GB our Best Buy award.

The Editor's choice goes to Seagate's Barracuda 7200.11 750 GB drives. To put it in simple words, the 750 GB makes more sense, given the quantity of data that resides on our hard drives—especially, HD movies and games. For us—1 TB drives are still expensive, but the 750 GB are sweetly priced! ■

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